

Hochschule
Bonn-Rhein-Sieg
University of Applied Sciences

b-it Bonn-Aachen International Center for Information Technology

Master's Thesis

Transformers-based Engagement Estimation: An Investigation of Fine-Tuning Strategies

Oviya Rajavel

Submitted to Hochschule Bonn-Rhein-Sieg,
Department of Computer Science
in partial fullfilment of the requirements for the degree
of Master of Science in Autonomous Systems

Supervised by

Prof. Dr. Teena Chakkalayil Hassan
Prof. Dr. Javad Ghofrani
Michał Stolarz, M.Sc
Mohammad Wasil, M.Sc

December 2026

© 2020 Oviya Rajavel. All Rights Reserved

[If AI assistants have not been used, use this sentence] I, the undersigned below, declare that this work has not previously been submitted to this or any other university and that it is, unless otherwise stated, entirely my own work.

[If an AI assistant has been used, use this sentence] I, the undersigned below, declare that this work has not previously been submitted to this or any other university and that it is, unless otherwise stated, entirely my own work. The report was, in part, written with the help of the AI assistant [AI assistant name] as described in the appendix. I am aware that content generated by AI systems is no substitute for careful scientific work, which is why all AI-generated content has been critically reviewed by me, and I take full responsibility for it.

Date	Oviya Rajavel

Abstract

Your abstract

Acknowledgements

Thanks to ....

Contents

1 Introduction **1**

1.1 Motivation . . . 2

1.1.1 2

1.1.2 2

1.2 Challenges and Difficulties . . . 2

1.2.1 2

1.2.2 2

1.2.3 2

1.3 Problem Statement . . . 2

1.3.1 2

1.3.2 2

1.3.3 2

2 State of the Art **3**

2.1 Background . . . 3

2.1.1 t-SNE . . . 3

2.2 Limitations of previous work . . . 3

3 Methodology **5**

3.1 Setup . . . 5

3.2 Adapters . . . 5

3.2.1 Structure . . . 5

3.2.2 Setup . . . 7

3.3 Experimental Design . . . 7

4 Solution **9**

4.1 Proposed algorithm . . . 9

4.2 Implementation details . . . 9

5 Evaluation and Results **11**

5.1 Experiment Description . . . 11

5.2 Experimental Setup . . . 11

5.3 Results . . . 11

6 Conclusions **13**
6.1 Contributions 13
6.2 Lessons learned 13
6.3 Future work 13

Appendix A Design Details **15**

Appendix B Parameters **17**

Appendix C [in the case of using AI-assistants] Description of AI-Generated Content **19**

References **21**

1

Introduction

User engagement (UE), or engagement, denotes the extent of a user's interest, attention, and active involvement in a given task. For example, in a classroom setting, a teacher can assess a student's engagement by observing behavioral cues such as attentiveness and expressed interest. In contrast, in a scenario where a robot serves as the instructor, the system may initially presume that the student is engaged [1] and proceed with a predefined teaching strategy. However, such an assumption can result in suboptimal learning outcomes if the student's actual engagement level differs from this expectation. Hence, the robot needs to determine whether the student is engaged or not to enhance teaching efficiency.

For engagement estimation, previous work has employed a variety of approaches, including dynamic Bayesian networks, traditional machine learning techniques, and deep learning models. Recent studies indicate that transformer-based methods have achieved improved performance in this domain [2].

Transformers with their attention mechanism revolutionized sequential processing by enabling the models to capture long-range and global dependencies between inputs and outputs [3]. Prior to the introduction of the transformer architecture, recurrent neural networks (RNNs) and their variants were widely used for sequential modeling but these architectures often struggled with long-range dependencies. Transformers address these limitations by employing self-attention mechanisms [3]. Following their remarkable success in the field of Natural Language Processing (NLP), transformer architectures have been successfully extended to other domains, including computer vision, speech processing, and multi-modal learning [4].

In the domain of computer vision, ViT (Vision Transformer) a transformer-based architecture for images, was released in 2020 which has shown better performance than the convolutional neural networks and was able to overcome the inductive bias in the CNN [4]. Inductive bias refers to the pre-training assumptions, such as weight sharing (this bias assumes that the same feature detectors are useful throughout the image), local receptive fields (this bias is based on the belief that local features can represent meaningful patterns). Extending transformer architectures to video data has led to the development of Video Transformers (VT), which are designed to model both spatial and temporal information. VT model global dependencies by establishing all-to-all interactions across both spatial and temporal dimensions. This property enables them to effectively capture motion patterns, long-range temporal relationships, and dynamic variations in appearance throughout video sequences [5].

In this project, we are focusing on ViT and VT for engagement estimation for the use case of

robot-assisted language learning.

1.1 Motivation

1.1.1 ...

1.1.2 ...

1.2 Challenges and Difficulties

1.2.1 ...

1.2.2 ...

1.2.3 ...

1.3 Problem Statement

1.3.1 ...

1.3.2 ...

1.3.3 ...

2

State of the Art

2.1 Background

2.1.1 t-SNE

- t-SNE (t-distributed Stochastic Neighbor Embedding) used to visualize high dimensional data in a low-dimensional space using unsupervised non-linear dimensionality reduction
- Non-linear dimensionality reduction means that the algorithm allows us to separate data that cannot be separated by a straight line.
- it focuses on preserving the pairwise similarities between data points in a lower-dimensional space
- Uses gausian kernel to calculate the similarity (probability) between the selected pair of points
- Main hyper paramter is perplexity it decides how many neighbours should each point pay attention to. low perplexity focus on local structure,high perplexity focus on broader neighborhoods
- t-SNE takes the neighborhood relationships in a high-dimensional dataset, converts them into probabilities, creates a 2D/3D probability distribution using a heavy-tailed t-distribution, and uses gradient descent to make the two distributions as similar as possible.

Use as many sections as you need in your related work to group content into logical groups

Don't forget to correctly cite your sources [6].

2.2 Limitations of previous work

3

Methodology

- The selected learning rate is 2e-4 (paper), rank selected is 8, alpha is 16
- cosine learning rate schduler (paper)
- AdamW optimizer
- batch_size = 4
- gradient_accumulation_steps = 4 (so bath size is 16) (balances the slightly higher lr)
- Included warmup steps for inital phase to overcome LR from shooting up the begining

3.1 Setup

3.2 Adapters

In this project we are using AdaptFormer [7], as one of the PEFT finetuning methods. In their work, Chen et al. propose an adapter architecture specifically designed for Vision Transformers (ViTs), targeting both image and video recognition tasks.The proposed approach is evaluated using ViT-Base models, with the image-based model pretrained on ImageNet-21K and the video-based model pretrained using VideoMAE.

According to the experimental results presented in the paper, AdaptFormer, through its proposed AdaptMLP modules, improves the transferability of Vision Transformers compared to full fine-tuning and prompt tuning, while requiring only approximately 2% of the total model parameters to be trained [7]. By restricting fine-tuning to a small number of additional parameters, the method also helps mitigate catastrophic forgetting.

3.2.1 Structure

Each Transformer encoder block primarily consists of two sublayers: a Multi-Head Self-Attention (MHSA) layer and a Multi-Layer Perceptron (MLP) layer. AdaptFormer introduces an additional lightweight adaptation branch alongside the original MLP, forming a modified MLP structure referred to as AdaptMLP.The AdaptMLP consists of two parallel branches. The first branch retains the original MLP

structure, preserving the pretrained representations, while the second branch introduces a lightweight adapter module with a bottleneck architecture and a nonlinear activation function.

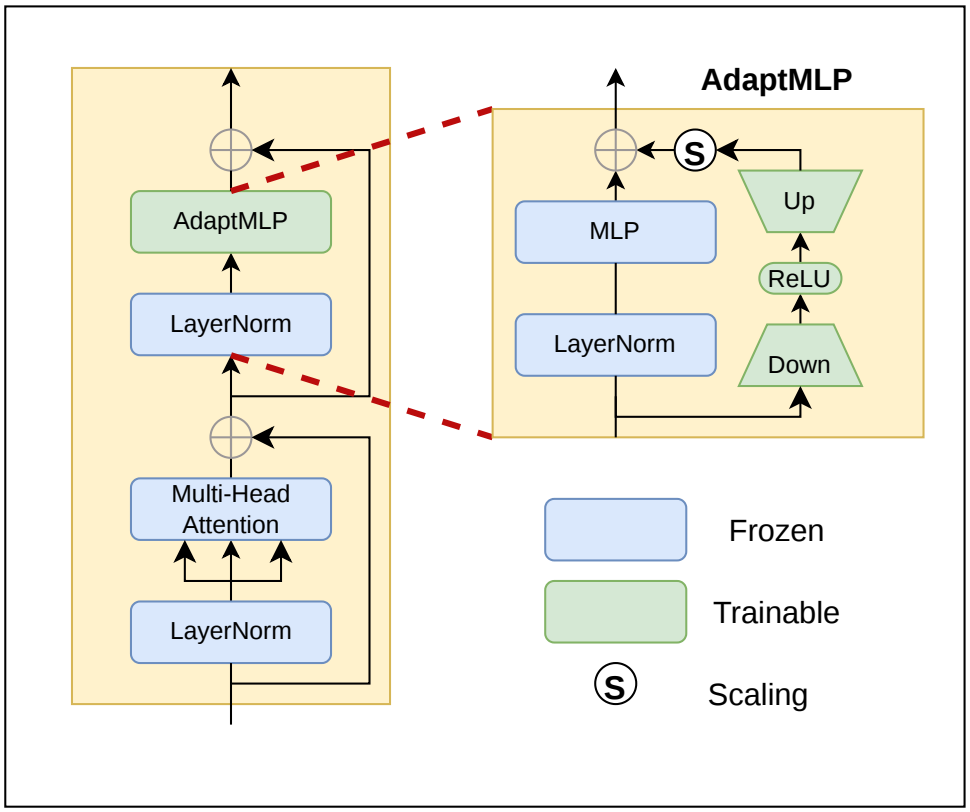


Figure 3.1: Architecture of the AdaptMLP module.

As illustrated in Figure 3.1, the adapter branch consists of three main components: a trainable down-projection layer, a ReLU activation function, and a trainable up-projection layer. The down-projection reduces the dimensionality of the input features, while the up-projection maps the resulting representation back to the original feature dimension.

Let d denote the feature dimension and $\hat{d}$ the bottleneck dimension, where $\hat{d} \ll d$. For a given input feature representation x_0^l, the adapter branch generates an adapted feature representation $\tilde{x}^l$, which can be expressed as:

$$\tilde{x}^l = \mathrm{ReLU}\left(\mathrm{LN}(x_0^l)W_{\mathrm{down}}\right)W_{\mathrm{up}}.$$

Here, LN denotes layer normalization, while W_{down} and W_{up} represent the trainable down-projection and up-projection weight matrices, respectively.

Finally, the output of the original MLP branch and the adapted features are combined through residual

connections with a scaling factor s. The resulting output is given by:

$$x^l = \text{MLP}\left(\text{LN}(x_0^l)\right) + s \cdot \tilde{x}^l + x_0^l.$$

The first term represents the output of the original MLP, which retains the feature transformations learned during pretraining. The second term represents the adapted features produced by the lightweight adapter module, weighted by a scaling factor s. This factor controls the influence of the adapter on the resulting feature representation. Finally, the original input x_0^l is added through a residual connection, preserving the input information and facilitating stable feature propagation. During fine-tuning, only the parameters of the adapter branch are updated, while the pretrained MLP and the remaining backbone parameters are kept frozen.

3.2.2 Setup

To ensure a consistent adaptation strategy across vision models with different hidden dimensions, we adopt a fixed bottleneck reduction ratio of 12 rather than a fixed bottleneck dimension. Accordingly, the bottleneck dimension is determined proportionally to the hidden dimension of each backbone. This choice maintains a consistent relative adapter capacity across architectures while allowing the number of adapter parameters to scale with the size of the underlying model. Although AdaptFormer uses a bottleneck dimension of 64 in its default experimental configuration, this corresponds to a reduction factor of 12 for a ViT-Base model with a hidden dimension of 768. We therefore generalize the relative bottleneck configuration to the different vision backbones considered in this study.

3.3 Experimental Design

4

Solution

Your main contributions go here

4.1 Proposed algorithm

4.2 Implementation details

5

Evaluation and Results

5.1 Experiment Description

Describe the experiments/evaluation you are performing to analyse your method.

5.2 Experimental Setup

Describe your experimental setup in detail.

5.3 Results

Describe the results of your experiments in detail.

6

Conclusions

6.1 Contributions

6.2 Lessons learned

6.3 Future work

A

Design Details

Your first appendix

B

Parameters

Your second chapter appendix

C

[in the case of using AI-assistants] Description of AI-Generated Content

Describe in detail what content was generated using an AI assistant. Name the AI assistant and how it was used (e.g. which prompts were used, and for which parts of the project).

References

[1] Kai-Yi Chin, Zeng-Wei Hong, and Yen-Lin Chen. Impact of using an educational robot-based learning system on students' motivation in elementary education. *IEEE Transactions on learning technologies*, 7(4):333–345, 2014.

[2] Sandeep Mandia, Kuldeep Singh, Rajendra Mitharwal, Faisel Mushtaq, and Dimpal Janu. Transformer-driven modeling of variable frequency features for classifying student engagement in online learning. *arXiv preprint arXiv:2502.10813*, 2025.

[3] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.

[4] Sonain Jamil, Md Jalil Piran, and Oh-Jin Kwon. A comprehensive survey of transformers for computer vision. *Drones*, 7(5):287, 2023.

[5] Salman Khan, Muzammal Naseer, Munawar Hayat, Syed Waqas Zamir, Fahad Shahbaz Khan, and Mubarak Shah. Transformers in vision: A survey. *ACM computing surveys (CSUR)*, 54(10s):1–41, 2022.

[6] Author Name. Book title. *Lecture Notes in Autonomous System*, 1001:900–921, 2003.

[7] Shoufa Chen, Chongjian Ge, Zhan Tong, Jiangliu Wang, Yibing Song, Jue Wang, and Ping Luo. Adaptformer: Adapting vision transformers for scalable visual recognition. *Advances in neural information processing systems*, 35:16664–16678, 2022.